Loan Approval Analysis

CodeTrade AI/ML Internship · Phase 1 Capstone

1. Question & Why It Matters

Which applicant characteristics are associated with loan approval?

Loan approval is important for both banks and applicants. This analysis explores how factors such as Credit History, Applicant Income, Education, and Property Area are related to loan approval. Understanding these relationships can help identify patterns in the loan approval process.

2. Data & Cleaning

The dataset contains 614 loan applications and 13 columns. It includes applicant details, income, loan information, and loan status.

The data was checked for missing values, incorrect data types, and unusual values. Cleaning was performed to prepare the dataset for analysis and improve the reliability of the results.

3. What I Did

I cleaned and explored the dataset using Python, pandas, NumPy, and data visualisation techniques. I compared approved and rejected applications using charts, summary statistics, and statistical tests.

I also examined relationships between Applicant Income, LoanAmount, Education, Credit History, and Loan Status.

4. What I Found

- Credit History: Applicants with a Credit History value of 1 had a higher approval rate than applicants with a value of 0. This shows

a strong association between Credit History and loan approval in this dataset.

- Applicant Income: The comparison of income between approved and rejected applicants did not provide sufficient evidence of a difference in their average incomes.
- Education: Among graduates, the median income was 3,812 for both approved and rejected applications. Among non-graduates, the median income was 3,378 for approved applications and 3,279 for rejected applications. These differences were relatively small.
- Income and Loan Amount: Applicant Income and LoanAmount had a positive correlation of 0.303. This indicates that applicants with higher incomes tended to have higher loan amounts.
- Other Relationships: Most other numerical variables showed weak relationships with one another.

5. What It Means

The analysis suggests that Credit History was strongly associated with loan approval in this dataset. Applicant Income alone did not provide sufficient evidence of a difference in average income between approved and rejected applicants.

Considering multiple applicant characteristics can help provide a clearer understanding of loan approval patterns. However, these findings should not be used as the only basis for making loan decisions.

6. Limitations

- **Missing Credit History values:** Missing values in `Credit_History` were filled with `0`. Using the mode or another imputation method could have changed the results of the Credit History analysis and its observed association with loan approval.
- **Outlier treatment:** Potential outliers in `ApplicantIncome`, `CoapplicantIncome`, and `LoanAmount` were identified

using the IQR method and replaced with median values. This may have affected the distributions and statistical results.

- **Dataset limitations:** The dataset contains 614 loan applications and may not represent all applicants, banks, or lending practices.
- **Statistical limitations:** The analysis identifies associations and differences but cannot establish that any applicant characteristic causes loan approval.
- **Group size:** Differences in the number of observations across applicant groups may affect the reliability of some comparisons.
- **Unobserved factors:** The dataset may not include every factor used in real-world loan approval decisions.

Conclusion

This analysis explored the relationships between applicant characteristics and loan approval. Credit History showed a strong association with approval, while Applicant Income alone did not provide sufficient evidence of a difference in average income between approved and rejected applicants.

Applicant Income and LoanAmount also showed a positive relationship, while income differences within education groups were small. Overall, the analysis identifies useful patterns in the dataset, but additional information would be needed to explain loan approval decisions more fully.